How Demanding Is Synth?

Adapting TimesFM-3 to a Live Forecasting Competition

Synth Internal review copy

Abstract. How much task-specific work does a strong pretrained forecaster need to compete on Synth? We study TimesFM-3 on the 24-hour cryptocurrency task and compare it with our in-house Chronos-2 miner. With published weights and a baseline conversion to price trajectories, TimesFM-3 ranks 224th of 257 and 81st of 252 on two tuning periods. Five cumulative adaptations improve those ranks to 3rd of 257 and 1st of 252. The best TimesFM-3 configuration on the two tuning periods has a median end-of-period rank of 19th of 257 across five evaluation periods, against 1st of 257 for Chronos-2. At those endpoints, TimesFM-3's smoothed score is 0.4–18.2% higher than Chronos-2's. In the three-day diagnostics, Chronos-2 has lower CRPS on every asset at the 5- and 30-minute horizons; results for the 24-hour price are mixed across assets. Synth is demanding in two respects: published weights require substantial task adaptation, and reaching the top of the ranking on tuning does not ensure comparable results in later periods. Rank and relative score differences show both competitive placement and the gap that remains.

1 Introduction

Public benchmarks compare forecasting models on established datasets. Synth asks a different question: how well does a forecasting system compete on subsequent financial observations against miners developed for that task?

Synth is a decentralized forecasting competition on Bittensor subnet 50. Participants, called *miners*, submit probabilistic price forecasts. Validators request forecasts and compare them with subsequent prices. A published scoring rule determines each miner's rank and reward weight. The competition covers several assets and forecast horizons [9].

We study Crypto 24h on BTC, ETH, SOL, XRP and HYPE: 1,000 trajectories over 288 five-minute steps per request (Figure 1). The continuous ranked probability score (CRPS) evaluates 5-minute, 30-minute, 3-hour and 24-hour forecasts. It rewards predictions that are precise and well calibrated about their uncertainty; lower is better [1, 9].

We adapt TimesFM-3 [6] and compare it with our in-house Chronos-2 miner [4]: a new entrant built around a published model versus an established miner. Two periods are used for adaptation and five further periods for evaluation.

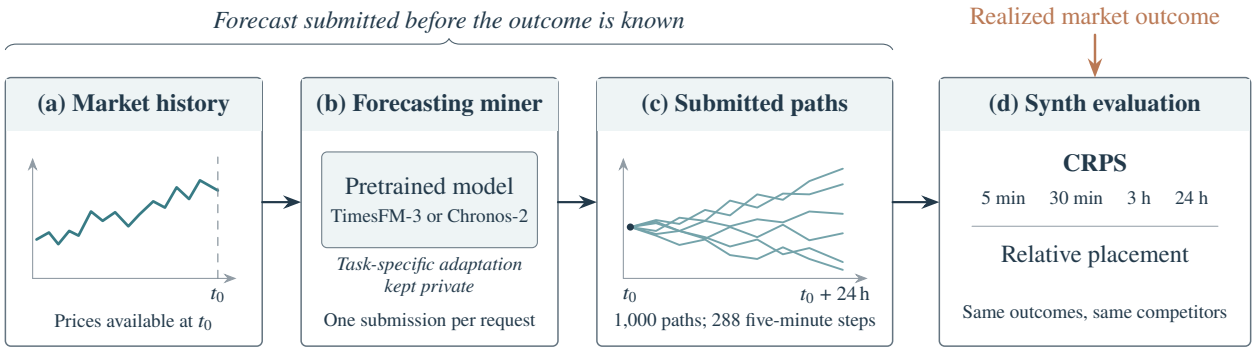


Figure 1. From market history to a Synth submission. The miner forecasts trajectories, which the validator evaluates at several intervals. The traces illustrate the workflow, not measured predictions.

Key result

Five adaptations bring the cumulative TimesFM-3 configuration into the top five on both tuning periods. The best TimesFM-3 tuning configuration nevertheless has a median end-of-period rank of 19th of 257 on evaluation, against 1st of 257 for our Chronos-2 miner. A competitive tuning result is therefore a starting point, not evidence that the remaining gap has been closed.

2 Background and related work

TimesFM [2] and Chronos-2 [4] are pretrained time-series models. Both families support multivariate and covariate-informed forecasting. Google's release evaluation ranks TimesFM-3 ahead of Chronos-2 on fev-bench, TIME and GIFT-Eval [6]. Those results compare pretrained models, not our adapted systems.

Shared benchmarks support reproducible comparisons [5], but repeated evaluation can make their tasks a target for model design and tuning. This risk of benchmark overfitting, or *benchmaxing*, relates to adaptive holdout reuse [7]. Pretraining overlap and limited statistical coverage further complicate forecasting evaluation [8].

Synth complements these tests with forecasts submitted before their outcomes are observed, against competitors that can change. Historical replays let us examine the adaptation needed for this task; prospective performance remains a separate question.

3 The miners

3.1 TimesFM-3: from quantiles to trajectories

Our TimesFM-3 candidate uses the 330.7M-parameter model with five-minute log returns as input.[1] The model predicts nine quantiles per step, from the 10th to the 90th percentile [3]. Knowing the likely price range at each future hour does not determine whether a morning fall is followed by a rebound or a further fall. Those alternatives can produce different distributions of price changes over longer intervals. A trajectory generator must therefore specify how values at successive steps are connected. The outer quantiles also leave a choice about the tails.

The baseline converts TimesFM-3's quantile outputs into sampled price trajectories. Its results evaluate this conversion together with the published weights, not a ready-made miner supplied by Google. We then add Distribution Calibration, Path Rearrangement, Signal Enrichment, Memory Enhancement and Market Specialization in a fixed order. Market Specialization includes task-specific fine-tuning. Section 5 reports the best tuning configuration carried into evaluation. Implementation details remain private.

3.2 Our in-house Chronos-2 miner

Our in-house miner uses the 119.5M-parameter Chronos-2 model, whose 21-quantile output also requires conversion into trajectories [4]. Its configuration reflects months of work on Synth and remains private. We compare this fixed, adapted miner with TimesFM-3, rather than claiming a controlled comparison of the two pretrained models.

4 Experimental protocol

4.1 Periods and configuration selection

Fine-tuning data end no later than April 1, 2026 for both models. The two tuning periods are used to compare input choices and trajectory-generation settings, rather than to fit model weights on those dates. We then carry the best TimesFM-3 tuning configuration into five further evaluation periods. Table 1 gives the three-day scoring periods used for evaluation and CRPS diagnostics; forecasts are issued one day earlier. The diagnostic sample contains 707 tuning and 1,800 evaluation forecasts per system.

We compute daily realized volatility from five-minute log returns, normalize by each asset's median over April 1–September 10, 2026, and combine assets using validator weights. The period index averages the three issuance days. It describes the sampled market conditions, rather than establishing that either model faces an out-of-distribution task.

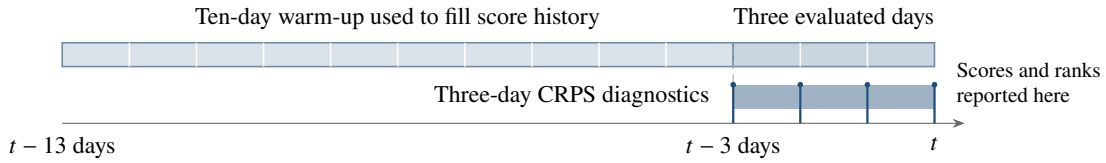
Table 1. Tuning and evaluation periods (scoring dates in 2026). Volatility describes the corresponding three issuance days.

Role	Period	Volatility
Tuning	May 28–30	1.16
Tuning	June 12–14	1.13
Evaluation	July 12–14	0.80
Evaluation	August 15–17	0.43
Evaluation	August 21–23	2.35
Evaluation	August 28–30	1.19
Evaluation	September 5–7	0.96

[1] Google distributes TimesFM-3 under the TimesFM Non-Commercial License v1.0 [11]. TimesFM-3 was used for research evaluation only; no TimesFM-3-based miner was deployed and no weights or derivatives were distributed.

4.2 Evaluation metrics

We replay the validator through `synth-lib` [9, 10]. Ten warm-up days fill the scoring history before each three-day evaluation period; only observations within that period are retained. Throughout this paper, *score* means the validator's smoothed score over the rolling ten-day history ending at each observation.



The three evaluated days follow a ten-day warm-up. Each cell represents one day.

4.2.1 Competition rank

Rank orders the smoothed scores at each retained timestamp; lower is better. Figures 2 and 3 show competitive placement. TimesFM-3 and Chronos-2 are inserted separately, so both can rank first without equal scores. The reported median uses the five end-of-period evaluation ranks.

4.2.2 Relative score difference

Let $S_{\mathrm{T}}(\tau)$ and $S_{\mathrm{C}}(\tau)$ denote the smoothed scores of the TimesFM-3 candidate and our in-house Chronos-2 miner at the same retained timestamp τ. Lower is better. Their relative difference quantifies gaps that rank may hide:

$$G_{\mathrm{score}}(\tau) = 100\left(\frac{S_{\mathrm{T}}(\tau)}{S_{\mathrm{C}}(\tau)} - 1\right). \tag{1}$$

Positive values favor Chronos-2; negative values favor TimesFM-3. Figures 2–3 and Table 3 retain the supplied end-of-period observations, not three-day averages.

4.2.3 Median per-request rank

A request's rank is one plus the number of recorded competitors with strictly lower CRPS. Table 2 reports the median by asset on tuning: performance on a typical request, rather than placement by smoothed score.

5 Results

5.1 What task adaptation changes

At the end of the tuning periods, the TimesFM-3 baseline ranks 224th of 257 in May and 81st of 252 in June. Five cumulative adaptations bring it to 3rd of 257 and 1st of 252 (Figure 2). Our in-house Chronos-2 miner ranks first in both periods and has a lower score in both. Task adaptation changes the practical competitiveness of the published model substantially, but the cumulative comparison does not isolate each stage's effect.

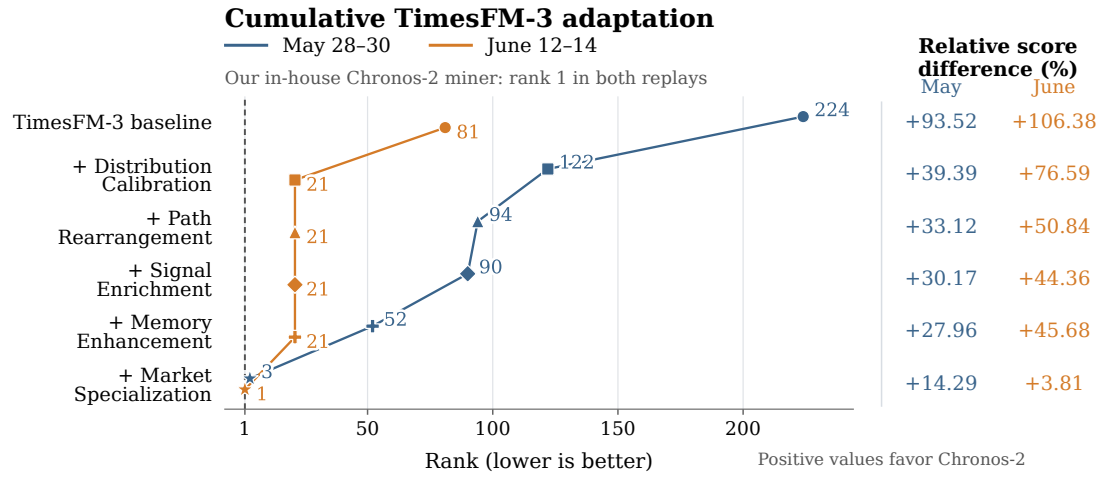


Figure 2. Cumulative adaptation on the two tuning periods. Marker shapes identify stages. The columns give each configuration's relative validator-score difference from Chronos-2; positive values favor Chronos-2, which ranks first in both periods.

June's four intermediate configurations all rank 21st despite different relative score gaps (Figure 2). The fourth stage slightly worsens the score without changing rank. In May, the first addition moves TimesFM-3 from 224th to 122nd. These changes reflect uneven spacing between competitors' scores: an improvement changes rank only when it crosses another competitor's score.

For evaluation, we retain the best configuration on the two tuning periods. It ranks 3rd of 257 in May and 1st of 252 in June. Figure 3 and Table 3 report results for this retained configuration.

Table 2. Median per-request ranks by asset on tuning (approximately 142 requests per asset). These rank the CRPS of individual requests, not the smoothed scores used for the competition rank.

Configuration	BTC	ETH	SOL	XRP	HYPE
Our Chronos-2 miner	44	28	29	45	87
TimesFM-3, released weights	191	145	150	171	169
+ Distribution Calibration	139	136	122	129	124
+ Path Rearrangement	126	129	112	106	144
+ Signal Enrichment	108	101	107	102	129
+ Memory Enhancement	121	79	100	85	113
+ Market Specialization	36	17	31	34	79

The final cumulative TimesFM-3 configuration has better median per-request ranks on four of the five assets (Table 2). These medians summarize individual requests in the three-day samples. Smoothed scores preserve score magnitudes and use a rolling ten-day history; the two measures need not order the systems in the same way.

5.2 Rank and relative score difference on evaluation

Our in-house Chronos-2 miner has a lower score at the end of all five evaluation periods and both tuning periods. Median end-of-period evaluation ranks are 1st of 257 for Chronos-2 and 19th of 257 for TimesFM-3. Both attain 1st of 257 in their separate September replays (Figure 3).

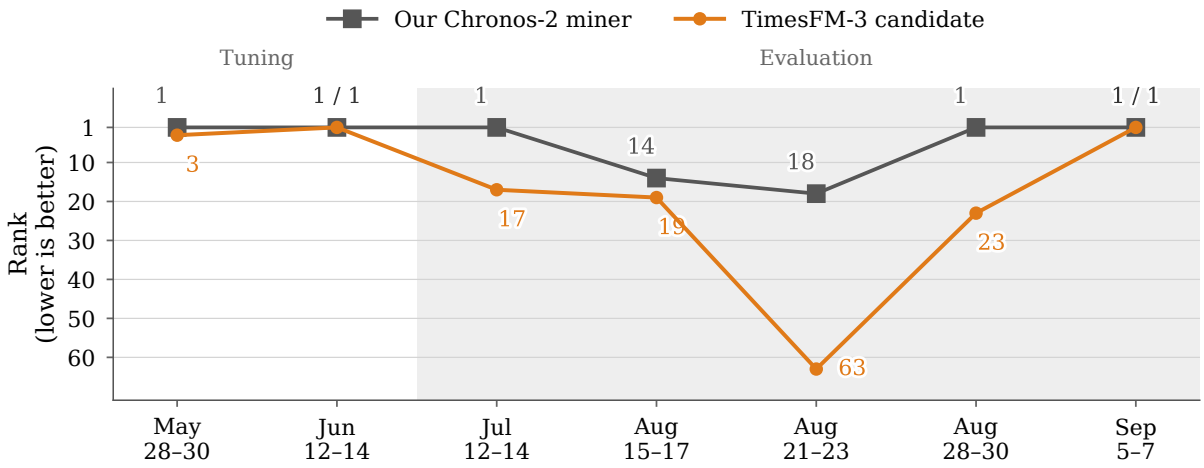


Figure 3. Ranks of the fixed TimesFM-3 candidate and our Chronos-2 miner at the end of the labelled three-day periods. Shading identifies evaluation periods. Points retain the supplied observations; lines connect periods, not a continuous rank history.

Table 3 reports the relative difference between the smoothed scores at the end of each period. TimesFM-3's score is 0.4–18.2% higher at the five evaluation endpoints, favoring our Chronos-2 miner in each comparison. The smallest gap occurs in September, when both systems rank first in separate replays. A shared rank therefore does not imply equal forecasting scores.

A larger relative score gap need not imply a worse rank: TimesFM-3 is 3rd of 257 in May at 14.3%, but 63rd of 257 on August 21–23 at 14.5%. The latter period also has the highest three-day volatility. This coincidence does not establish a volatility effect.

Table 3. Relative difference between the smoothed scores of the TimesFM-3 evaluation candidate and our Chronos-2 miner. Positive values favor Chronos-2. Rows retain the supplied end-of-period observations within the labelled three-day windows; they are not three-day averages. Ranks appear in Figure 3.

Role	Period	Relative score difference (%)
Tuning	May 28–30	+14.3
Tuning	Jun 12–14	+3.8
Evaluation	Jul 12–14	+11.6
Evaluation	Aug 15–17	+16.5
Evaluation	Aug 21–23	+14.5
Evaluation	Aug 28–30	+18.2
Evaluation	Sep 5–7	+0.4

6 Forecast-quality diagnostics

6.1 Fine-tuning and trajectory quality

The component comparisons identify two complementary areas of work: adapting the model to financial data and improving the trajectories generated from its outputs. Fine-tuning improves the forecasts, but the conversion from quantiles to paths remains important after training. Input selection contributes to this process: useful market information can improve the forecasts, while adding features does not systematically help a fine-tuned model. These results support developing the input representation and trajectory generator alongside the model weights, rather than treating path generation as a final formatting step.

6.2 CRPS by asset and horizon

The horizon analysis compares raw CRPS rather than the validator score. For each asset and horizon, the relative CRPS gap is $100(C_T/C_C - 1)$, where C_T and C_C are sums over the same diagnostic requests. Positive values favor Chronos-2.

Figure 4 covers the five three-day evaluation samples, with 360 requests per asset. Chronos-2 has lower CRPS in all ten cells at 5 and 30 minutes. Their relative gaps are below 0.6%. At 3 hours, TimesFM-3 has lower CRPS only on SOL.

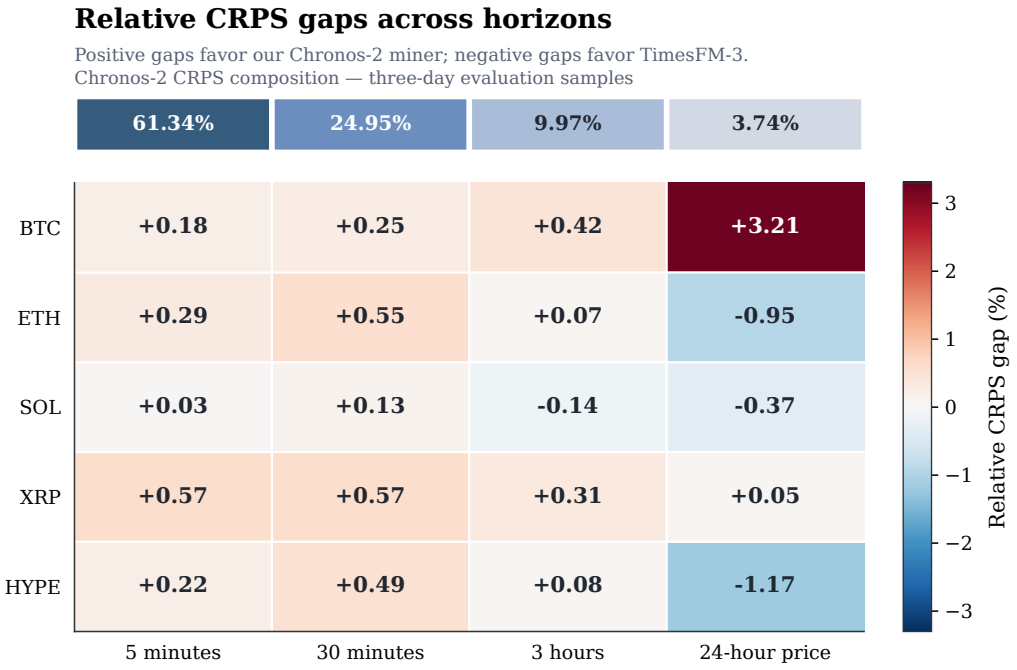


Figure 4. Relative CRPS gaps on the five three-day evaluation samples. Positive values favor Chronos-2. Header percentages are each horizon's share of Chronos-2's summed CRPS over 1,800 forecasts. Values are percentages; the color scale is symmetric.

For the 24-hour price, TimesFM-3 has lower CRPS on ETH, SOL and HYPE, but its BTC gap is +3.21%. The validator-weighted average of the five asset-level gaps is +0.523%, favoring Chronos-2. Three asset-level wins are therefore not an aggregate advantage. These estimates do not establish statistical significance or equivalence.

The shorter intervals contribute 61.34% and 24.95% of Chronos-2's diagnostic CRPS: 86.29% together. The protocol sums CRPS contributions from 288 five-minute, 48 thirty-minute and eight three-hour increments, plus the 24-hour price. This structure favors numerous short-horizon terms, without fixing their percentage shares. These raw-CRPS contributions are distinct from the relative differences between smoothed scores.

7 Discussion and limitations

Task adaptation brings TimesFM-3 close to the top on tuning, but the fixed TimesFM-3 candidate remains behind our Chronos-2 miner at the evaluation endpoints reported here. Rank shows where TimesFM-3 stands among competitors; the relative score difference shows the gap to our Chronos-2 miner even when their ranks coincide. These are complementary views of the task-specific work still needed in the evaluated periods.

This retrospective comparison covers five evaluation periods and one fine-tuning seed. The two systems differ in their training data and inference procedures, so the results compare the evaluated forecasting systems rather than the pretrained models alone.

8 A benchmark for participants

The public `synth-lib` backtester computes CRPS, scores, ranks and reward weights for participants' forecasts [10]. From the repository at the cited revision, after installing dependencies and generating prediction files:

```
RUN=synth_lib/backtester/scripts
uv run "$RUN/run_backtest.py" \
  --miner-name my_model \
  --competition crypto-24h \
  --asset ALL --days 20
```

Here, `--days 20` provides a ten-day warm-up followed by roughly ten days of ranked results; each score still uses a ten-day rolling window. The example retains a longer observation period than the three days used here. Prediction files and market data must cover the requested history. If no prediction files are found, the runner falls back to random-walk forecasts.

Three directions are worth testing: time-dependent trajectories from quantiles, fewer severe forecast failures and missed submissions, and training against multi-horizon CRPS. Our Chronos-2 results provide a comparison for participants developing these alternatives.

References

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